

IILM UNIVERSITY, GREATER NOIDA

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INTERNSHIP REPORT

AI-ML VIRTUAL INTERNSHIP

Submitted in partial fulfillment of the academic requirements

Submitted By	Mohd Muzammil Akhtar
Roll No.	2410031716
Year / Section	3rd Year, 3CSE25
University	IILM University, Greater Noida
Internship Duration	8 Weeks (June – August 2026)

Academic Year: 2026

AI-ML Virtual Internship Report



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MINISTRY OF
EDUCATION



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During June - August 2026

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GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

CANDIDATE’S DECLARATION

I, **Mohd Muzammil Akhtar**, Roll No. **2410031716**, student of **3rd Year, 3CSE25**, IILM University, Greater Noida, hereby declare that this internship report entitled “**AI-ML Virtual Internship**” is a record of the learning and practical work completed by me during the 8 Weeks (June – August 2026).

I have prepared this report for academic submission and have presented the concepts, tools, workflows, observations and learning outcomes in my own words. Wherever external documentation or learning resources were consulted, they are acknowledged in the bibliography/references section.

I further declare that the information presented in this report is intended to document the internship learning experience and does not claim ownership of third-party software, datasets, frameworks, logos or institutional material.

Student Name	Mohd Muzammil Akhtar
Roll No.	2410031716
Year / Section	3rd Year, 3CSE25
Signature	_____
Date	_____

ACKNOWLEDGMENT

I would like to express my sincere gratitude to IILM University for providing the academic environment and guidance required to undertake this internship. The internship gave me an opportunity to connect classroom concepts with practical applications in Artificial Intelligence and Machine Learning.

I am thankful to the internship ecosystem and learning resources made available through the AICTE–EduSkills Virtual Internship programme and its associated industry-supported learning initiatives. The structured exposure to AI/ML concepts, model development, evaluation and responsible use of data helped me understand the workflow followed in real-world machine-learning tasks.

I also thank my faculty members, mentors, classmates and everyone who supported me during the internship. Finally, I am grateful for the opportunity to complete this eight-week learning experience and present its outcomes in this report.

Mohd Muzammil Akhtar

Roll No.: 2410031716

3rd Year, 3CSE25

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are widely used to build systems that can learn patterns from data and support prediction, classification, recommendation and decision-making. The AI-ML Virtual Internship provided a structured learning environment for understanding these ideas through a practical, step-by-step workflow.

The internship focused on the complete machine-learning lifecycle: understanding a problem, collecting or inspecting data, preparing features, selecting an appropriate model, training it, evaluating its performance and communicating the result. Equal importance was given to data quality, reproducibility, interpretation and responsible use of AI.

This report documents the internship learning experience, the technologies studied, the methodology followed, the architecture of a typical AI/ML solution and the expected outcomes. It is written at an academic level so that the workflow can also be reused for future CSE projects.

4.2 Organization / Programme Profile

The internship certificate identifies the programme as an AICTE–EduSkills Virtual Internship for 2026. The certificate also indicates support from the India Edu Program and Google for Developers. For this report, the programme is treated as a structured virtual learning and practical-training experience in Artificial Intelligence and Machine Learning.

The learning model combines guided study with hands-on experimentation. Such an approach is useful for a computer-science student because it develops both conceptual understanding and the ability to translate a problem statement into an executable data-driven workflow.

Programme focus: Artificial Intelligence, Machine Learning, data preparation, model building, model evaluation, practical experimentation and responsible AI awareness.

4.3 Problem Statement

Many real-world problems involve large amounts of data and require a systematic way to discover patterns or make predictions. Manual analysis can

become slow, inconsistent and difficult to scale. The problem addressed through the internship was therefore to understand how an AI/ML workflow can convert raw or structured data into a useful predictive or analytical output.

A complete solution must do more than train a model. It should begin with a clear objective, use suitable data, handle missing or inconsistent values, select meaningful features, avoid data leakage, train the model correctly, evaluate it with suitable metrics and communicate limitations.

4.4 Project Objectives

No.	Objective
1	Understand the fundamentals of Artificial Intelligence and Machine Learning.
2	Learn the standard machine-learning lifecycle from problem definition to evaluation.
3	Practice data preprocessing, feature preparation and exploratory analysis.
4	Understand supervised-learning algorithms and when to use them.
5	Train, validate and evaluate a model using appropriate performance metrics.
6	Develop awareness of overfitting, underfitting, bias, data leakage and responsible AI.
7	Improve documentation, presentation and technical communication skills.

4.5 Scope of the Project

The scope covers the educational and practical stages of an AI/ML project rather than a single commercial deployment. It includes problem understanding, data preparation, exploratory analysis, model selection, training, evaluation, documentation and presentation.

Included: supervised-learning concepts, basic unsupervised-learning awareness, Python-based data handling, feature preparation, model training, evaluation metrics, visualization and responsible-AI considerations.

Not included: production-scale cloud deployment, proprietary enterprise datasets, high-risk automated decision systems, or claims of a specific production accuracy without a verified dataset and experiment record.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Programming language for data handling, experimentation and ML workflows.

Technology / Tool	Purpose
Jupyter Notebook / similar notebook environment	Interactive experimentation, explanation and visualization.
NumPy	Numerical arrays and mathematical operations.
Pandas	Data loading, cleaning, transformation and tabular analysis.
Matplotlib	Charts and visual exploration of data and model results.
Scikit-learn	Preprocessing, model training, validation and evaluation.
Git / version control concepts	Tracking changes and maintaining reproducible work.

The exact software environment may vary depending on the learning platform and notebook setup. The important point is the workflow: data is prepared in a controlled way, the model is trained on training data, and its performance is checked on data that was not used to fit the model.

4.7 System Architecture

A typical AI/ML solution can be represented as a pipeline. Data enters the system, is validated and prepared, transformed into useful features, passed to a learning algorithm, evaluated and finally presented as a prediction or analytical result.

Problem Definition Define target and success criteria	Data Collect / inspect relevant data	Preprocessing Clean, transform and prepare features
Model Training Fit algorithm on training data	Validation Tune and compare alternatives	Evaluation Measure performance on unseen data
Interpretation Understand results and limitations	Deployment / Use Use predictions appropriately	Monitoring Check drift and future performance

Figure 4.1: AI/ML learning workflow

4.7.1 Input Layer

The input layer contains the problem statement and the available data. A good AI/ML project starts by identifying what must be predicted or classified, what information is available, and what output would be useful to a user.

4.7.2 Data Processing Layer

This layer handles missing values, duplicate records, inconsistent formats, categorical variables, numerical scaling and feature selection. Data preprocessing is critical because a model can learn incorrect patterns if the input data is inconsistent or biased.

4.7.3 Model Layer

The model layer contains the learning algorithm. For classification, algorithms such as Logistic Regression, Decision Tree, Random Forest or Support Vector Machine can be considered. For regression, Linear Regression, Decision Tree Regression or Random Forest Regression may be appropriate depending on the problem.

4.7.4 Evaluation and Output Layer

The final layers measure how well the trained model performs and present the result. Accuracy alone is not always sufficient; precision, recall, F1-score, confusion matrix, mean absolute error or root mean squared error may be selected according to the task.

4.8 Methodology

The internship methodology follows a repeatable machine-learning lifecycle. The process is designed to reduce errors and make experiments easier to explain and reproduce.

Step 1 – Problem Understanding

Define the objective, identify the target variable, understand the intended user and select a measurable success criterion.

Step 2 – Data Understanding

Inspect the dataset structure, data types, class distribution, missing values, duplicates and unusual observations.

Step 3 – Data Preprocessing

Clean data, encode categorical variables, scale numerical features when needed, and split data into training and testing sets.

Step 4 – Exploratory Data Analysis

Use descriptive statistics and visualizations to understand relationships, distributions and possible data-quality issues.

Step 5 – Feature Preparation

Select or construct useful input variables while avoiding information that would leak the answer into the training process.

Step 6 – Model Selection

Choose baseline and candidate algorithms based on the task, dataset size, interpretability and expected performance.

Step 7 – Training

Fit the model using training data and tune parameters using an appropriate validation strategy.

Step 8 – Evaluation

Evaluate the final candidate on unseen data and inspect task-specific metrics and error patterns.

Step 9 – Interpretation

Explain what the model has learned, where it performs well, where it fails and what limitations remain.

Step 10 – Documentation

Record assumptions, preprocessing steps, model choices, metrics and observations so the work can be reviewed or reproduced.

Figure 4.2: Model development cycle



4.8.1 Model Evaluation

Model evaluation should match the problem type. In classification, a confusion matrix shows the number of correct and incorrect predictions across classes. Precision indicates how many predicted positive cases were actually positive, while recall indicates how many actual positive cases were identified. The F1-score balances precision and recall.

For regression, common measures include Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). The choice of metric should reflect the practical cost of different errors.

Metric	Main Use	Interpretation
Accuracy	Classification	Fraction of total predictions that are correct.
Precision	Classification	Reliability of positive predictions.
Recall	Classification	Ability to find actual positive cases.
F1-score	Classification	Balance between precision and recall.
MAE	Regression	Average absolute prediction error.
RMSE	Regression	Penalizes larger errors more strongly.

Table 4.1: Common model-evaluation measures

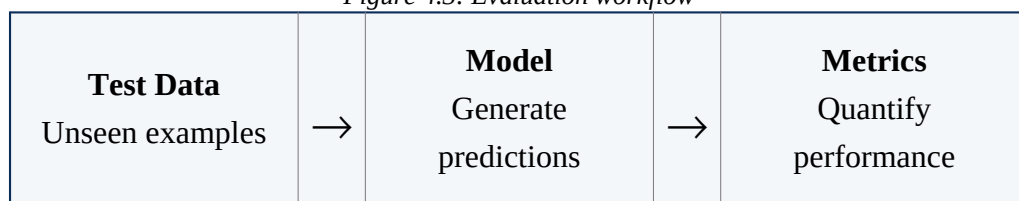
4.8.2 Avoiding Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on new data. Underfitting occurs when the model is too simple to capture useful patterns. These problems can be studied by comparing training and validation performance and by using suitable model complexity, regularization or cross-validation.

4.8.3 Data Leakage and Responsible AI

Data leakage happens when information that would not be available at prediction time enters the training process. It can make evaluation look unrealistically strong. Responsible AI also requires attention to privacy, fairness, transparency, security and the consequences of incorrect predictions.

Figure 4.3: Evaluation workflow



4.8.4 Practical Learning Activities

1. Understanding AI, ML, deep-learning terminology and common application areas.
2. Working with structured data and learning the role of rows, columns, features and targets.
3. Practicing data cleaning and transformation before model training.
4. Using visualizations to identify distributions, relationships and possible anomalies.
5. Understanding the difference between training, validation and testing data.
6. Comparing simple baseline models with more complex candidate models.
7. Studying evaluation metrics and interpreting a confusion matrix for classification.
8. Understanding the causes of overfitting, underfitting and data leakage.
9. Learning how responsible-AI principles affect data collection, model selection and deployment.
10. Preparing technical documentation and presentation material to communicate results.

4.8.5 Skills Developed

Skill Area	Learning Developed
Programming	Python fundamentals for data and ML experimentation.
Data Handling	Cleaning, transformation, feature preparation and exploratory analysis.
Machine Learning	Understanding training, validation, model selection and evaluation.
Problem Solving	Breaking a broad problem into measurable and testable steps.
Documentation	Writing structured technical explanations and maintaining clear records.
Communication	Presenting AI/ML concepts and results in an understandable manner.

4.9 Expected Outcomes

1. A clear understanding of the end-to-end AI/ML development lifecycle.
2. Ability to explain the role of data quality and preprocessing in model performance.

3. Ability to select basic machine-learning approaches according to the problem type.
4. Ability to interpret common evaluation metrics instead of relying on a single score.
5. Awareness of overfitting, underfitting, leakage, bias and responsible AI.
6. Improved confidence in Python-based data analysis and machine-learning experimentation.
7. Improved technical writing, documentation and presentation skills.

4.10 Certificate of Completion and Other Proof



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Supporting Documentation

The internship completion certificate shown on the previous page is the formal proof of completion provided for this report. If the university requires additional communication/e-mail proof, the relevant screenshot or PDF can be inserted here without changing the chapter structure.

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The above references were used for general conceptual understanding and technical terminology. No claim is made that the cited books or documentation represent the student's internship work.